Population for Individuals amongst Species Program

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Date: 7/13/2026

Tier 1

The Problem:

Many people struggle to accurately identify wildlife species due to limited knowledge and the similarities between certain animals. This can make it more difficult to educate the public, increase awareness of biodiversity, and support wildlife conservation efforts. An AI-powered animal identification system can provide quick and accurate species identification while helping users learn more about wildlife.

The Solution

To address this problem, our application will use a computer vision model to identify wildlife species from uploaded images. The model will analyze each image, predict the animal species, and display the species name, confidence score, and educational information to help users identify and learn about wildlife.

Workflow

User uploads a wildlife image
⬇️
Image is preprocessed and prepared for analysis
⬇️
MobileNetV2 image classification model analyzes the image
⬇️
Model predicts the wildlife species
⬇️
Application displays the species name, confidence score, and educational information such as habitat, diet, and conservation status.

Technical Approach

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Computer Vision Technique: Image Classification

Model Architecture: Convolutional Neural Network (CNN)

Model: MobileNetV2

How We Will Use It: We will use transfer learning with a pretrained MobileNetV2 model to classify wildlife species from uploaded images.

Framework: TensorFlow and Keras using Google Colab.

Why This Approach?

We chose image classification because our application only needs to identify one wildlife species from each uploaded image. MobileNetV2 is a lightweight, pretrained CNN that performs well with transfer learning and can be trained efficiently using Google Colab's free resources. TensorFlow and Keras provide an open-source framework that makes it easy to build, train, and evaluate our model.

Data Plan

- Data Origin: Public
- Dataset Contains: Photos of Real Wildlife taken from Africa.
- Dataset Preparations: Adjust photos as needed for size, organize photos for efficiency, split into necessary metrics for proper measurements.

Success Metrics

Target Metric:

Primary Goal:

- The algorithm correctly analyzes the animal, and gives it a basic profile of said species. Percentage Target: 80%

Secondary Goals:

- Good Performance
- Attention to Detail
- Little to no sign or clear indication of Bias

Milestones

Week
Milestone
Goal

Week 5: (MILESTONE) Blueprint: (GOAL): Complete the proposal slides, create the GitHub repository, and submit the project plan.

Week 6: (MILESTONE): First Working Demo: (GOAL): Test a pretrained MobileNetV2 model with sample wildlife images to make sure the application works.

Weeks 7–8: (MILESTONE): Build the Application: (GOAL): Train the model using our wildlife dataset and develop the wildlife identification application.

Week 9: (MILESTONE): Improve & Measure: (GOAL): Test the application, improve accuracy, and evaluate the model against our success metrics.

Week 10: (MILESTONE): Final Presentation: (GOAL): Finalize the application, update the GitHub repository and documentation, and present the completed project.

Risks and Resources

Risks:

- The Algorithm develops confirmation bias, causing it to misidentify and forces a hard reset on its programming
- The algorithm fails to get anything correctly.
- The algorithm hits goal of identification, but fails to render any meaningful information as a result.

Resources:

- Python
- Collab
- YOLO